

Replicability of Two Classic Experiments in Social Psychology: Implicit Association Test and Imagined Contact

Yinqi Huang

Abstract

This research project examined the replicability of two classic social psychology effects: intergroup bias on a gender-based Implicit Association Test (IAT) and reduced intergroup bias through "Imagined Contact." We hypothesized that interaction between study participants and cross-gender experimenters before the experiment would exaggerate the implicit gender stereotypes captured by the IAT. We also hypothesized such interaction would amplify the positive effects of "Imagined Contact" on reducing the stereotypes. The experimental results replicate the original "Imagined Contact" study. The results also show that the "Imagined Contact" measured by the explicit attitudes test are affected by the interaction between study participants and cross-gender experimenters. Specifically, when study participants and experimenters were of cross gender, they showed less gender biases in explicit measures. They also showed a less activation of the gender-science bias when they were of the same gender. Our results suggest that the gender of experimenter influences participants' responses in explicit measures and Gender-Science IAT, though in different ways. To improve replicability of this study, we recommend that researchers report the experimenters' and participant's gender. Given the design of this study, it is unclear whether its low replicability was due to the gender of the experimenter or witnessing a social interaction similar to the explicit measures setup. To further partition the effect of these two factors, we recommend conducting follow-up studies. We also observed a dissociation between explicit measures and implicit measures, which was likely a result of participants wanting to maintain a positive self-image by avoiding disclosure in the explicit measures.

Keywords: replication crisis, intergroup contact, gender, Implicit Association Test (IAT), imagined contact

Replicability is an important predictor of the validity of an experiment. A study with high replicability allows independent researchers to generate the same findings when adopting the procedure, materials, and sample population of the original study (Shrout & Rodgers, 2018). Replication crisis refers to the widespread observation that many psychological studies suffer from low replicability (Klein, et al., 2014; Klein, et al., 2018; Ebersole, et al., 2016). This is a serious issue in social psychology as it raises concerns over research ethics and significance of psychological findings (Maxwell, Lau, & Howard, 2015; Shrout & Rodgers, 2018). However, a failure to replicate does not mean that we should disregard the original study entirely: many factors contribute to low replicability rates, such as restrictions related to a specific experimental condition, misinterpretation of statistics, the emergence of new findings, and the development of new theories, to name a few (Shrout & Rodgers, 2018). In this study, we were

interested in replicating the results of the "Imagined Contact" study conducted by Husnu and Crisp and revalidating the Implicit Association Test (IAT).

Imagined contact occurs when participants imagine interacting with a member from a specific social group. The original "Imagined Contact" study conducted by Husnu and Crisp (2010) showed that imagining cross-group contact reduced explicit stereotypes. Researchers invited 33 non-Muslim undergraduate students to participate in the experiment. The participants were asked to imagine either a neutral scenario or engaging in a social interaction with a Muslim. The results showed that students who were instructed to imagine an intergroup interaction were more willing to engage in future social contact with Muslims. Thus, "Imagined Contact" can be an effective tool in reducing intergroup stereotypes. However, the "Imagined Contact" study is replicated with a small effect size in the first

"Many Labs" project (Klein, et al., 2014).

The Implicit Association Test is developed for implicit attitudes measured by tapping into automatic association of social categories. Essentially, by comparing the reaction times of the pairing of members of two categories, researchers are able to measure the relative strength of automatic associations of the pairs and determine the dominant association (Greenwald, Nosek, & Banaji, 2003). Social psychologists rely heavily on this tool as existing literature argues that IAT possesses high validity and reliability (Greenwald, Nosek, & Banaji, 2003; Nosek, Greenwald, & Banaji, 2005; Greenwald, Poehlman, Uhlmann, & Banaji, 2009). However, most of the reviews exclude the potential mediating effect of the experimental environment, especially the influence of experimenters who directly engage with the study participants. Previous studies discovered that environmental factors can prime a participant's performance. For example, non-linguistic cues such as national flags and pictures in the experimental settings can prime bilinguals and influence their accessibility of languages or their knowledge of the tested culture (Jared, Poh, & Paivio, 2012; Grainger, Declerck, & Marzouki, 2017). Even though psychologists take into account experimenter effects when designing experiments, sometimes the nature of the tasks makes it impossible to control for. For example, if the task is designed to measure social interaction, the conventional approaches such as double blind are ineffective as the experimenters have to interact with the participants to deliver the tasks, either through in person contact or via computer. This interaction can still prime participants responding to IAT in a certain way despite both parties having no prior knowledge of the hypothesis.

The objective was to investigate the moderating role of environmental settings on these effects. We addressed two research questions in this study: (1) Would mere exposure to the target group prior to the assessments (i.e., through a same- or cross-gender experimenter) increase the accessibility of gender stereotypes in IAT and thus amplify the implicit bias measured by IAT? (2) Would the exposure to a similar scenario prior to the assessments increase the

accessibility of the target scenario in the "Imagined Contact" task and thus amplify its effects on reducing gender bias?

Pilot Study

We conducted a pilot study survey two months prior to the main experiment with the research assistants at the Social Psychophysiological Research & Quantitative Methods Laboratory (SPRQL). The purpose of this survey was to decide which characteristics we should focus on in the main experiment.

Method

Participants and Procedure

Nine research assistants who worked at SPRQL from September 2018 to December 2018 submitted their responses anonymously. There was one question in the survey: What aspects would stand out the most when you are engaging with a study participant? The survey was shared with all the research assistants internally by the lab manager using Google Forms. The full survey can be found in the materials section on the project website (<https://osf.io/v8aq7/>).

Results

The final sample consists of nine observations. The results show that sex ($n = 6$), race ($n = 3$), and age ($n = 3$) are the three most salient characteristics of a study participant in the context of a professional laboratory setting. Refer to Table 1 for the full results. Due to the time restriction of this project, we focused on sex, the most noticeable characteristics, in the main experiment. Since all participants and experimenters reported themselves as cisgender, to match gender performance captured by Gender IAT and the "Imagined Contact" task, we focused on the *gender* of the study participants and the *gender* of the experimenters.

Main Experiment

The "Imagined Contact" study (husnu & Crisp, 2010) suggests that prior contact with a social group reduces future intergroup biases.

We hypothesized that: (1) participants run by an cross-gender experimenter would have a smaller gender bias compared to participants run by a same-gender experimenter; (2) participants run by a cross-gender experimenter and witness a cross-gender conversation would have a smaller gender bias compared to participants run by a same-gender experimenter and do not witness a conversation; and (3) participants who were assigned to the "Imagined Contact" condition would have a smaller gender bias compared to the control condition.

We designed three "Experimenter Conditions" (independent variable 1) to address the three hypotheses: (C1) gender-matched experimenter condition with scenario 1 (S1); (C2) gender-mismatched experimenter with S1; (C3) gender-mismatched experimenter with scenario 2 (S2). There were also two scenario settings (independent variable 2): (S1) one experimenter interacted with the study participant; and (S2) one experimenter, one female confederate, and one male confederate interacted with the study participant. In S2, the two confederates engaged in conversation in the hallway, which acted to prime the participants for the questions in the "Imagined Contact" task. The matching/mismatching of gender was then crossed with the two scenarios settings. In C1 and C2, if the participants agreed to participate, they were escorted to the experiment room. In C3, the participants waited in the hallway for five minutes while two confederates engaged in a conversation on a neutral topic within proximity of the participant. In contrast to C2, we introduced an environmental component: whether the participant witnessed the interaction. This component was added to exaggerate the environmental effect by adding more intergroup interaction related factors. We were curious about how experiencing the interaction would differ from witnessing it vicariously.

Method

Participants

Twenty-three undergraduate students (14 females, 9 males) were recruited using an internal recruitment system PSYNUP. All participants provided consent for using their data. Due to

limited access to personal information, we were only able to match the sex of the researchers and that of the participants prior to the experiment. Nonetheless, all participants agreed that their gender identity aligned with their sex at the end of the test. Thus, the gender performance measured in this study was aligned their declared sex.

Materials

Explicit gender attitudes. The first half of this task was an exact replication of Husnu and Crisp's (2010) study. The objects in the scenario scripts were changed to incorporate the student experiment at the University of Toronto. We also included a "Feeling Thermometer" task at the end of the "Imagined Contact" task as an explicit gender attitude measure. The task was conducted on the Qualtrics survey platform. The full text is available on the project website <https://osf.io/ygmub/>.

Implicit gender attitudes. We included all the gender IATs available in the Inquisit library. Inquisit is a provider of psychological tests and experimental paradigms. The two available gender IATs were Gender-Career IAT and Gender-Science IAT. The IAT scripts are available at <https://osf.io/3zegh/>.

Procedure

Participants were recruited and randomly assigned to one of the three "Experimenter Conditions" mentioned above. On the experiment day, participants were welcomed by their experimenter, who would go through the consent process with them in a private room. The hypothesis was masked from the participants in the consent form to avoid priming. In C1 and C2, if participants decided to participate, they were escorted to the experiment room. In C3, the participants waited in the hallway for five minutes while two confederates engaged in a conversation on a neutral topic within proximity of the participant. After being escorted to the experiment room, the participants were seated in front of a computer, where they would complete two gender-based IATs and one "Imagined Contact" task. The order of IATs and the

"Imagined Contact" task were counterbalanced between participants. After completing the three tasks, the participants filled out a manipulation check questionnaire online. Upon completion, the experimenter returned to the room, presented a debriefing form to the participants, and revealed the study hypothesis. The participants were given an opportunity to ask questions about the experiment and withdraw their data from the study at that time.

Data Analysis

Before proceeding to the ANOVA analysis, we performed a normality test on the data. We then examined mean differences in the three types of gender prejudice (i.e., explicit gender attitudes, implicit association between gender and family vs. career, implicit association between gender and arts vs. science) as a function of the "Experimenter Condition" (3: Same-gender Experimenter, Cross-gender Experimenter with No Witnessed Cross-gender Interaction, Cross-gender Experimenter with a Witnessed Cross-gender Interaction) with a series of three ANOVAs, one for each dependent variable. The explicit prejudice model included an additional "Imagined Contact" manipulation (2: Control, "Imagined Contact").

All analyses controlled for the gender of participants. For any statistically significant effects, we conducted post-hoc, pairwise analyses to compare their means, adjusting for multiple comparisons using Tukey's adjustment of the p -value. We calculated the partial effect sizes for all predictors in the models using partial η^2 . We interpreted these effect sizes using the conventions set by Cohen (1992), where partial $\eta^2 \geq 0.26$ was interpreted as "large," values including 0.13 up to 0.26 were considered "medium," and partial η^2 values including 0.02 up to 0.13 were considered "small." Values where $\eta^2 < 0.02$ were considered trivial.

Results

Normality Test

The Shapiro-Wilk Test failed to detect non-normality in the data. For Explicit Gender

Attitudes data, $p = .588$; for Gender-Career IAT data, $p = .509$; for Gender-Science IAT data, $p = .504$. Thus, our data was considered as normally distributed.

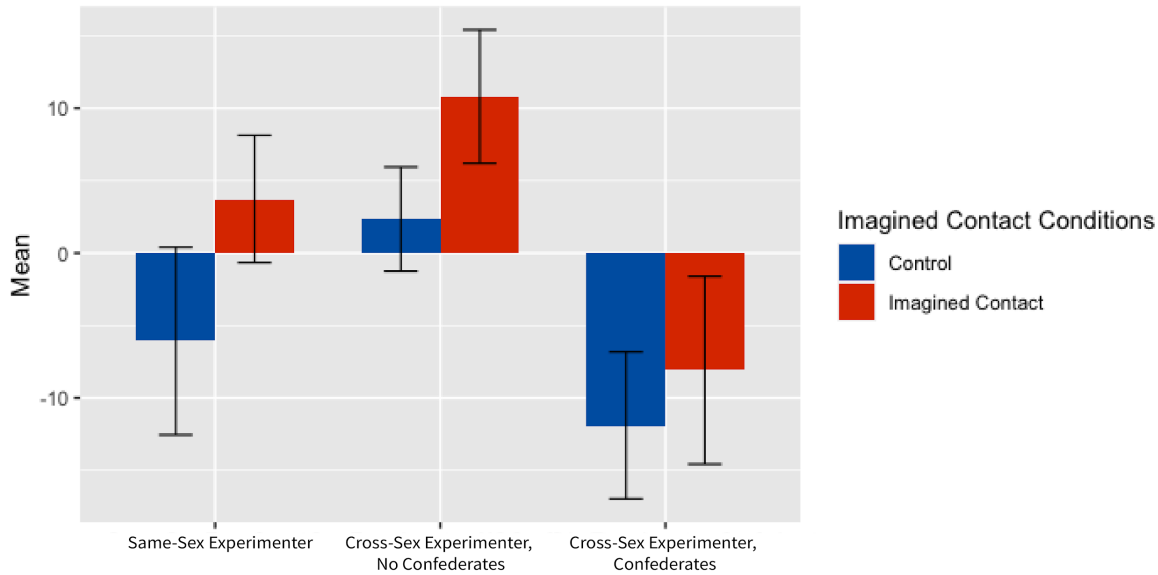
Explicit Gender Attitudes

Pro-male gender bias was modelled as a function of the "Experimenter Condition," the "Imagined Contact" condition, and the interaction of the "Experimenter Condition" and the "Imagined Contact," controlling for participant gender. As shown in Figure 1, there was a large main effect of the "Experimenter Condition", $F(2, 14) = 4.91, p = .024$, partial $\eta^2 = .41$. Post-hoc analyses revealed this effect was driven by the difference between the two conditions that had cross-gender experimenters but differed in whether a cross-gender interaction was vicariously witnessed, $t(14) = 3.13, p = .019$. Specifically, participants whose session was run by a cross-gender experimenter and who witnessed a cross-gender interaction in the lab showed less explicit gender bias ($M = -9.99, SE = 4.16$) than participants whose session was run by a cross-gender experimenter and did not witness a cross-gender interaction ($M = 6.57, SE = 2.96$). There was a medium-sized but non-significant main effect of "Imagined Contact" condition, $F(1, 14) = 3.11, p = .100$, partial $\eta^2 = .18$. The interaction between the "Experimenter Condition" and the "Imagined Contact" condition was insignificant, $F(2, 14) = 0.15, p = .863$, partial $\eta^2 = .02$. The effect of participant gender was trivial, $F(1, 14) = 0.002, p = .967$, partial $\eta^2 < .001$.

Gender-Career IAT

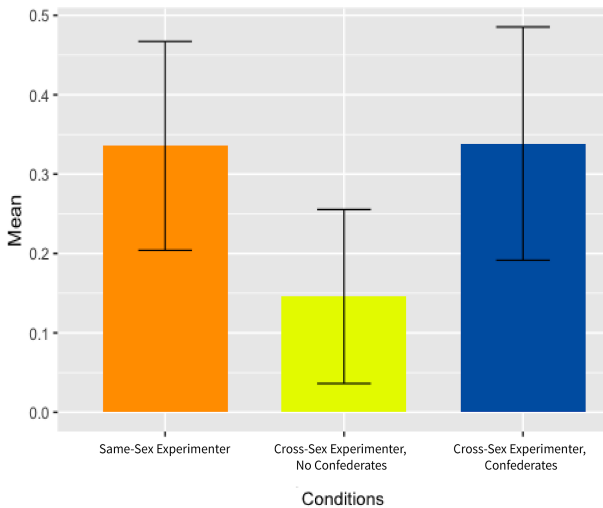
Traditional implicit attitudes about gender and careers were modelled as a function of the "Experimenter Condition," controlling for participant gender. As shown in Figure 2, there was no main effect of the "Experimenter Condition," $F(2, 17) = 0.72, p = .50$, partial $\eta^2 = .078$. Moreover, there was no main effect of participant gender, $F(1, 17) = 0.18, p = .68$, partial $\eta^2 = .01$.

Figure 1. Comparison of Explicit Gender Attitudes Responses of Control Group vs. Imagined Contacted Group in the Three Experimenter Conditions



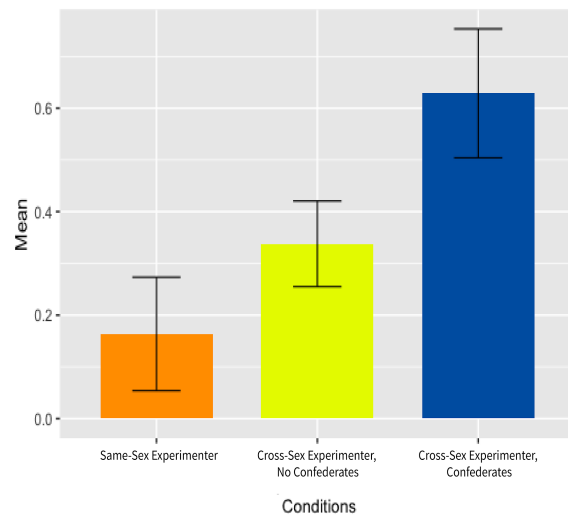
Note. The y axis Mean refers to Estimated Marginal Means, and the x axis refers to the three “Experimenter Conditions”: 1) Same-sex experimenter with no confederate, 2) Cross-sex experimenter with no confederate, and 3) Cross-sex experimenter with two confederates. The error bar represents the standard error of Estimated Marginal Means

Figure 2. Comparison of Gender-Career IAT in the three Experimenter Conditions



Note. The y axis Mean refers to Estimated Marginal Means, and the x axis refers to the three “Experimenter Conditions.” The error bar represents the standard error of Estimated Marginal Means.

Figure 3. Comparison of Gender-Science IAT in the three Experimenter Conditions



Note. The y axis Mean refers to Estimated Marginal Means, and the x axis refers to the three “Experimenter Conditions.” The error bar represents the standard error of Estimated Marginal Means.

Gender-Science IAT

Traditional implicit attitudes about gender and science were modelled as a function of the "Experimenter Condition," controlling for participant gender. There was a large and statistically significant main effect of the "Experimenter Condition," $F(2, 15) = 3.88, p = .044$, partial $\eta^2 = .34$. As shown in Figure 3, participants ran by a same-gender experimenter showed less traditional implicit attitudes about gender and science ($M = 0.17, SE = 0.11$) than participants whose session was run by a cross-gender experimenter and witnessed a cross-gender interaction in the lab ($M = 0.64, SE = 0.13$), $t(15) = -2.79, p = .035$. For implicit beliefs about gender and science, there was a medium-sized main effect of participant gender, $F(1, 15) = 3.60, p = .077$, partial $\eta^2 = .19$.

Discussion

The goal of this study was to investigate (1) Would mere exposure to the target group prior to the assessments (i.e., through a same- or cross-gender experimenter) increase the accessibility of gender stereotypes in IAT and thus amplify the implicit bias measured by IAT? (2) Would the exposure to a similar scenario prior to the assessments increase the accessibility of the target scenario in the "Imagined Contact" task and thus amplify its effects on reducing gender bias?

The Explicit Gender Attitudes, Gender-Career IAT, and Gender-Science IAT were treated as three separate measurements in the main experiment for two reasons. First, the order of the three tests was counterbalanced and thus minimized the impact of the test order. Second, the results of implicit attitudes test and explicit attitudes test should be relatively constant despite the order and the number of items within each test according to a review conducted by Nosek, Greenwald, and Banaji (2005).

Explicit Gender Attitudes

The gender of experimenter had no impact on participants' responses, thus, the result does not support our first hypothesis. However, witnessing a cross-gender interaction decreased explicit gender biases, which supports the second hypothesis. The third hypothesis is supported by

the finding that participants who were assigned to the "Imagined Contact" condition had a smaller gender bias compared to the control condition. No interaction is observed between the "Imagined Contact" condition and the "Experimenter Condition."

Our results suggest that witnessing a cross-gender interaction vicariously is a stronger predictor of reduced bias than actually engaging in the interaction. This observation brings up an interesting question: how do these two factors differ from each other? One plausible explanation is "Demand Characteristics," a term defined by Orne (2000) as "the totality of cues and mutual expectations which inhere in a social context...which serve to influence the behaviour and/or self-reported experience of the research receiver." The cross-gender interaction participants witnessed a salient environmental cue, and it drew participants' attention to sex and/or gender. This hypothesis was supported by the results of the suspicion check survey (<https://osf.io/nr4zd/>), as a few participants believed that the study hypothesis was gender related. This observation might cause participants to monitor their responses. When there was a cue in the environment suggesting an expectation, participants with "Demand Characteristics" would modify their behavior to fulfill this expectation (Mccambridge, Bruin, & Witton, 2012). The awareness of the possible involvement of gender in the study hypothesis might cause participants to avoid disclosing their gendered attitudes (Nosek, 2005).

Previous studies have reported the environmental effects on our information processing system (Jared, Poh, & Paivio, 2012; Smith, et al., 2017; Grainger, Declerck, & Marzouki, 2017). For example, national flags could have a long-lasting impact on people's political attitudes (Carter, Ferguson, & Hassin, 2011). According to the attention probe theory proposed by Smith et al.(2017), witnessing cross-gender interaction activates gender-biased information processing, mediated through attention. In contrast, direct interaction with a cross-gender experimenter did not capture as much attention as witnessing a cross-gender interaction did, and thus had less impact on participants' responses.

There is a possible dissociation between the behavior of witnessing the interaction and the gender of the experimenter. Our studies only examined the combination of witnessing cross-gender interaction in the priming phase and imagining a cross-gender conversation in the "Imagined Contact" task. Follow-up studies are needed to rule out the possible involvement of gender by adopting different combinations of same-gender vs. cross-gender in the priming phase and in the "Imagined Contact" task.

Gender-Career IAT

Participants' performance in Gender-Career IAT did not vary in the "Experimenter Condition." One explanation is that the association between gender and career is strong and stable and thus is unaffected by the gender of the experimenter. However, we cannot conclude that Gender-Career IAT reflects the stereotypes that participants endorse. It could be that the experimenter's gender is a rather weak prime compared with more salient cues. One study showed that directly priming participants with the traditional labor division (male-career, female-family) increased the activation of traditional gender stereotypes (Rudman & Phelan, 2010). Thus experimenters' gender could be a weak prime compared with direct priming. And we cannot conclude that Gender-Career IAT scores were unaffected by other external factors.

Gender-Science IAT

Participants with a same-gender experimenter showed less automatic activation of the traditional gender-science association. This finding is surprising as it refutes the first and the second hypothesis. One plausible explanation is that the Gender-Science association was relatively flexible compared with the Gender-Career association and could be affected by external factors such as individual preference of topic (science vs. liberal arts) predicting Gender-Science IAT scores (Zitелny, Shalom, & Bar-Anan, 2017). The most significant difference is between participants with a same-gender experimenter and participants with a cross-gender experimenter and also had witnessed a cross-gender interaction. However, it is unclear whether it was driven by the gender of

experimenter, witnessing the interaction, or the combination of both variables. Follow-up studies are needed to examine whether witnessing a same-gender interaction would produce a similar result.

Conclusion

The goal of this study was to investigate the impact of exposure to environmental cues on the "Imagined Contact" effect and the accuracy of IAT measures. The pilot study showed that the participants' gender was the most observed trait to experimenters during testing. Thus, we decided to focus on the Explicit Gender Attitudes, Gender-Career IAT, and Gender-Science IAT in the main experiment.

Our results replicated the original "Imagined Contact" experiment conducted by Husnu and Crisp (2010) as participants in the "Imagined Contact" condition showed a reduced gender bias. We also noticed when study participants and experimenters were of cross gender, they showed less gender biases in explicit measures. They also showed a less activation of the gender-science bias when they were of the same gender. Our results suggest that the gender of experimenter influences participants' responses in explicit measures and Gender-Science IAT, though in different ways. To increase the replicability of studies that involve the use of explicit measures and Gender-Science IAT, we recommend that researchers report both experimenter's gender and participants' gender.

We also observed a discrepancy between the explicit measures and Gender-Science IAT. This was driven by the condition where the participant witnessed a cross-gender conversation and interacted with a cross-gender experimenter. Given the design of this study, we were not clear whether it was due to the gender of the experimenter or witnessing an interaction similar to the explicit measures setup. To further partition the effect of these two factors, we recommend conducting follow-up studies. Although previous studies have found a correlation between explicit attitudes measures and IAT (Greenwald, Poehlman, Uhlmann, & Banaji, 2009; Zitелny, Shalom, & Bar-Anan, 2017), we observed a dissociation among Explicit Gender Attitudes, Gender-Career IAT, and Gender-Science IAT.

Only the Gender-Career IAT score is independent of the "Experimenter Condition." The discrepancy between explicit measures and Gender-Science IAT could be the result of participants trying to maintain a positive self-image (Nosek, 2005). Self-presentation is one of four moderators of the discrepancy between explicit and implicit measures. Since people tend to present themselves in a positive, prosocial way in reporting negative responses, explicit measures fail to capture these responses. In our study, participants' awareness of the possible involvement of gender in the study hypothesis might cause them to avoid disclosing their gendered attitudes.

Unfortunately, the limitation of this study stems from its small sample size, and consequently, low statistical power. Thus, our results might also suffer from low replicability by overestimating the effect size (Button et al., 2013). Nonetheless, all study materials and raw data are available online and we welcome fellow researchers to verify our findings.

Acknowledgments

I gratefully acknowledge the support of my supervisor, Dr. Elizabeth Page-Gould, who contributed importantly to this project. Special thanks to Joel Le Forestier, Daniel An, and Helena Wang for their useful advice; to the lab managers, Bashir Fares and Tara Kendall, and the research assistants, Dorothy Sun, Gina Yan, Won Kyu Jung, Rayan Saied, and Terrence Zhang, for their help in data collection; and to Nesmith Chingcuanco, Stanley Leung, and Dylan Guan for their comments on my writing.

References

- Button, K. S., Ioannidis, J. P., Mokrysz, C., Nosek, B. A., Flint, J., Robinson, E. S., & Munafò, M. R. (2013). Power failure: Why small sample size undermines the reliability of neuroscience. *Nature Reviews Neuroscience*, 14(5), 365-376. doi:10.1038/nrn3475
- Carter, T. J., Ferguson, M. J., & Hassin, R. R. (2011). A single exposure to the american flag shifts support toward republicanism up to 8 months later. *Psychological Science*, 22(8), 1011-1018. doi:10.1177/0956797611414726
- Cohen, J. (1992). A power primer. *Psychological Bulletin*, 112(1), 155-159. doi:10.1037//0033-2909.112.1.155
- Ebersole, C. R., Atherton, O. E., Belanger, A. L., Skulborstad, H. M., Allen, J. M., Banks, J. B., . . . Nosek, B. A. (2016). Many labs 3: Evaluating participant pool quality across the academic semester via replication. *Journal of Experimental Social Psychology*, 67, 68-82. doi:10.1016/j.jesp.2015.10.012
- Grainger, J., Declerck, M., & Marzouki, Y. (2017). On national flags and language tags: Effects of flag-language congruency in bilingual word recognition. *Acta Psychologica*, 178, 12-17. doi:10.1016/j.actpsy.2017.05.004
- Greenwald, A. G., Nosek, B. A., & Banaji, M. R. (2003). Understanding and using the implicit association test: I. An improved scoring algorithm. *Journal of Personality and Social Psychology*, 85(2), 197-216. doi:10.1037/0022-3514.85.2.197
- Greenwald, A. G., Poehlman, T. A., Uhlmann, E. L., & Banaji, M. R. (2009). Understanding and using the Implicit Association Test: III. Meta-analysis of predictive validity. *Journal of Personality and Social Psychology*, 97(1), 17-41. doi:10.1037/a0015575
- Husnu, S., & Crisp, R. J. (2010). Elaboration enhances the imagined contact effect. *Journal of Experimental Social Psychology*, 46(6), 943-950. doi:10.1016/j.jesp.2010.05.014
- Jared, D., Poh, R. P., & Paivio, A. (2012). L1 and L2 picture naming in mandarin-english bilinguals: A test of bilingual dual coding theory. *Bilingualism: Language and Cognition*, 16(2), 383-396. doi:10.1017/s1366728912000685
- Klein, R. A., Ratliff, K. A., Vianello, M., Adams, R. B., Bahník, Š, Bernstein, M. J., . . . Nosek, B. A. (2014). Investigating variation in replicability. *Social Psychology*, 45(3), 142-152. doi:10.1027/1864-9335/a000178

- Klein, R. A., Vianello, M., Hasselman, F., Adams, B. G., Adams, R. B., Alper, S., . . . Nosek, B. A. (2018). Many labs 2: Investigating variation in replicability across samples and settings. *Advances in Methods and Practices in Psychological Science*, 1(4), 443-490. doi:10.1177/2515245918810225
- Maxwell, S. E., Lau, M. Y., & Howard, G. S. (2015). Is psychology suffering from a replication crisis? What does “failure to replicate” really mean? *American Psychologist*, 70(6), 487-498. doi:10.1037/a0039400
- Mccambridge, J., Bruin, M. D., & Witton, J. (2012). The effects of demand characteristics on research participant behaviours in non-laboratory settings: A systematic review. *PLoS ONE*, 7(6). doi:10.1371/journal.pone.0039116
- Nosek, B. A. (2005). Moderators of the relationship between implicit and explicit evaluation. *Journal of Experimental Psychology: General*, 134(4), 565-584. doi:10.1037/0096-3445.134.4.565
- Nosek, B. A., Greenwald, A. G., & Banaji, M. R. (2005). Understanding and using the implicit association test: ii. Method variables and construct validity. *Personality and Social Psychology Bulletin*, 31(2), 166-180. doi:10.1177/0146167204271418
- Orne, M., & Whitehouse, W. (2000). Demand characteristics. In *Encyclopedia of Psychology*. Washington, DC: American Psychological Association and Oxford University Press.
- Rudman, L. A., & Phelan, J. E. (2010). The effect of priming gender roles on women’s implicit gender beliefs and career aspirations. *Social Psychology*, 41(3), 192-202. doi:10.1027/1864-9335/a000027
- Shrout, P. E., & Rodgers, J. L. (2018). Psychology, science, and knowledge construction: broadening perspectives from the replication crisis. *Annual Review of Psychology*, 69(1), 487-510. doi:10.1146/annurev-psych-122216-011845
- Smith, L., Leung, W. G., Crane, B., Parkinson, B., Touloupoulou, T., & Yiend, J. (2017). Bilingual comparison of mandarin and english cognitive bias tasks. *Behavior Research Methods*, 50(1), 302-312. doi:10.3758/s13428-017-0871-0
- Zitелny, H., Shalom, M., & Bar-Anan, Y. (2017). What is the implicit gender-science stereotype? Exploring correlations between the gender-science IAT and self-report measures. *Social Psychological and Personality Science*, 8(7), 719-735. doi:10.1177/1948550616683017